

ViralFingerprinting_XiaoqiWu.pdf

by Yuki Wu

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Coursework Coversheet

Name	Xiaoqi Wu
Date	13/12/2023
Module	Genetics
Coursework item	Genetics Practical 1: Viral Fingerprinting

Self-reflection is a non-assessed part of the coursework submission to aid your learning

1. Have you used previous feedback/self-reflection to produce this coursework? If so, how? If not, why?

Yes, I did not manage my time correctly in the previous essay writing assignment. This time, I changed my plans and forced myself to complete it a few days before.

2. What do you feel went well? (e.g., development of specific skills, sense of achievement, part of the assignment you are most proud of)

I did a good job of time management in writing this essay by separating tasks into several days to complete it instead of finishing it at the last minute.

I read many literature and gained a lot of knowledge from this coursework.

I followed the instruction, using the format from the sample paper.

3. What do you feel did not go well? (e.g., learning outcomes that you found most difficult to meet, aspects of the assignment that were challenging, time-management)

I could practice more on paraphrasing sentences, which I found hard during writing the essay.

4. What did you hope to learn from doing this coursework? Did you achieve that?

I hope that I can learn more about this essay topic. I did gain much information from this activity by reading literature on this topic.

Exploring Viral Fingerprinting Potential: Insights from HERV-K (HML2) Family Analysis in Human Populations

Xiaoqi Wu
Department of Life Sciences,
Imperial College London,
London, United Kingdom

Abstract

This study targets insertionally polymorphic endogenous retroviruses (ERVs), specifically the HERV-K (HML2) family inherited from Neanderthals and Denisovans, the extinct archaic humans, aiming to explore the potential of viral fingerprinting by analysing the ERVs distribution among human populations. The investigation used polymerase chain reaction (PCR) and gel electrophoresis, and the presence of solo large terminal repeats (solo-LTR) and pre-integration sites were analysed. Accompanied by calculating distinct ERV combinations, findings contribute to exploring viral fingerprinting, and its advantages and disadvantages compared to microsatellite marker-based DNA fingerprinting have been discussed. This research provides insights into human population genetics and evolution.

Introduction

Endogenous retroviruses (ERVs) denote ancient germline infections wherein the integrated provirus becomes an integral part of the host genome, perpetually transmitted vertically from one host generation to the next (Boeke & Stoye, 1997; Gifford & Tristem, 2003). This human genome study focuses on insertionally polymorphic ERVs, particularly the HERV-K (HML2) family, which was introduced since the divergence of humans and chimpanzees (Donner et al., 1999). After being inherited from extinct archaic humans like Neanderthals and Denisovans, they are still active in modern human populations (Belshaw et al., 2004; Tristem, 2000). Employing Polymerase Chain Reaction (PCR) and gel electrophoresis, this research investigates ERV distribution, emphasising solo large terminal repeats (solo-LTR) differentiation among sub-populations, and unique ERV combinations. The study aims to provide valuable insights into viral fingerprinting, unravelling human population genetics and evolution complexities. Anticipated outcomes extend beyond evolutionary understanding to potential applications in medicine and forensic genetics, advancing molecular anthropology.

Methods

DNA isolation was performed using the Qiagen DNeasy Blood and Tissue Kit. Subsequently, PCR reactions were set up with the following components: sterile distilled water, 5X PCR reaction buffer, 1mM primer mix, 2mM stock nucleotide mix, DNA sample, 5U/ml Taq polymerase (Promega). These reactions underwent thermocycling and were then preserved by ethanol precipitation. Gel electrophoresis employed a 1.4% agarose gel stained with 'SYBR Gold.' Samples were loaded alongside a 50 ng/μl 1Kb plus ladder MWM from Invitrogen (6 μl). Electrophoresis was conducted at 80-100 Volts, monitored regularly, and concluded with gel photography.

Results

Specific primers are utilised in this study, targeting sequences flanking five solo-LTRs identified in Denisovan and Neanderthal genomes. A BLAST analysis confirmed the absence of alignment with the HERV-K family in humans, establishing that the observed sequences represented the dimensions of the preintegration site.

Solo-LTRs from the HERV-K family typically have an average length of 968 bp (Reus et al., 2001). Considering an additional six bases from duplicated cellular host DNA due to integrase activity and a 4-base reduction during the insertion process (Katz, Gravuer & Skalka, 1998), the calculated size of the preintegration site was determined (Table 1). The sizes of preintegration site products will be given as an output by entering the given primer set sequences into the UCSC *In silico* PCR program (Kent et al., 2002). It is inferred that products containing a solo-LTR inherited from Denisovan or Neanderthal ancestry should exhibit a length of 970 bases greater than the preintegration site.

Table 1 Product sizes of preintegration sites and solo-LTRs for corresponding primers. Ne and De mean solo-LTRs originally found in a Neanderthal genome and Denisovan genome.

Primer set	Primer set sequence	Expected sizes of pre-integration sites (bp)	Expected sizes of solo LTRs (bp)
Ne2	FOR 5'- CAGGAGGCTGAAATGGGAGG 3'	326	1300
	REV 5'- GCCACACTTTTTAGCATGGATG 3'		
De7	FOR 5'- GGGGAGAAATCAACCATCCATAAAC 3'	382	1356
	REV 5'- TGGACTTTGGGCACTGGGC 3'		
De9	FOR 5'- CCTCTGCTATGCCATACTGCC 3'	244	1218
	REV 5'- AACATATCAATTAACCGCATATTAAC 3'		
De10	FOR 5'- CTGCAGGTAACAGCTGGGC 3'	516	1488
	REV 5'- TAACTCAACCGGAAACGCGA 3'		
De11	FOR 5'- AGGCACTGGGGAATCCTACA 3'	669	1643
	REV 5'- TTGGCCAGAGAGTTGGAGC 3'		

The gel analysis of my genome sequences displays a single band in lane 2, measuring approximately 382 bp. This band is likely to be a preintegration site, as its size aligns with the values specified in Table 1, and there is no observable smearing above the band. In addition, a primer in lane 4 is designed to examine the presence of solo-LTR in lane 2, which shows no bands, affirming the absence of a solo-LTR in De7. No other preintegrated sites or solo-LTRs were detected, and bands below 100 bp were identified as primer dimers, subsequently excluded from consideration.

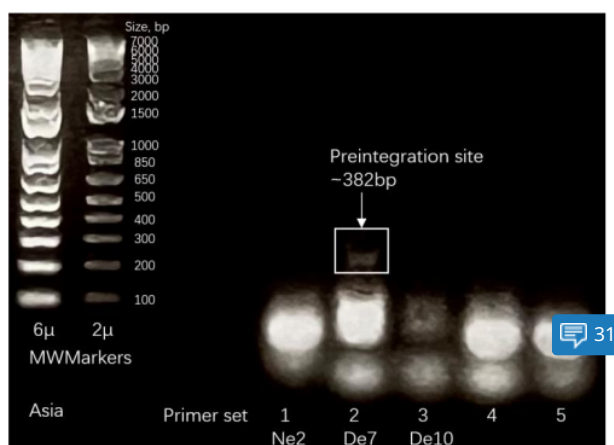


Fig. 1 Gel electrophoresis image testing the presence of solo-LTRs and preintegration sites.

Allele frequencies for solo-LTRs and preintegration sites were computed using five primer sets (Table 2), contributing to H_s and H_t calculations (Table 3). Sub-population De9 ERV data were excluded from subsequent analysis. F_{st} is an indicator for differentiation levels (Wright, 1978), highlighting De9 with low F_{st} ($0.0010 < 0.05$), suggesting minimal sub-population differentiation. Ne2 and De10 exhibit moderate differentiation ($0.05 < F_{st} < 0.15$), while De7 and De11 show substantial differentiation ($0.15 < F_{st} < 0.25$ and $F_{st} > 0.25$, respectively).

The Pearson chi-square statistics is used to assess Hardy-Weinberg Equilibrium (HWE). Null hypotheses were rejected for Ne2 and De7, indicating deviation from expected genotype frequencies ($\chi^2 > 3.841$). In contrast, De9, De10, and De11 remained in HWE, so no significant deviation exists (Table 4).

Table 2 Allele frequencies within the sub-population and for the entire population were calculated from the raw observed data. 'A' represents the allele of the solo-LTR, and 'a' represents the allele of the preintegration site. Sample calculations are exemplified: * $A = 8 \times 2 + 58 = 74$; ** $A \text{ allele} / (N_{\text{subpopulation}} \times 2) = 74 / (73 \times 2) = 0.507$; *** $A \text{ allele} / (N_{\text{subpopulation}} \times 2) = 104 / (99 \times 2) = 0.525$.

Primer	Asia	Africa	Europe, South America	Middle Eastern, North America, Oceania	Total
Ne2	AA	8	0	5	13
	Aa	58	0	19	78
	aa	7	0	1	8
	$N_{\text{subpopulation}}$	73	0	25	99
	A allele	74 *	0	29	104
	a allele	72	0	21	94
	A allele frequency (p)	0.507 **	/	0.580	0.525 ***
	a allele frequency (q)	0.493	/	0.420	0.475
De7	AA	9	0	1	10
	Aa	47	0	11	58
	aa	15	0	3	19
	$N_{\text{subpopulation}}$	71	0	15	87
	A allele	65	0	13	78
	a allele	77	0	17	96
	A allele frequency (p)	0.458	/	0.433	0.448
	a allele frequency (q)	0.542	/	0.567	0.552
De9	AA	4	0	0	4
	Aa	7	0	1	8
	aa	5	0	0	5
	$N_{\text{subpopulation}}$	16	0	1	17
	A allele	15	0	1	16
	a allele	17	0	1	18
	A allele frequency (p)	0.469	/	0.500	0.471
	a allele frequency (q)	0.531	/	0.500	0.529
De10	AA	2	0	3	5
	Aa	13	0	6	19
	aa	10	0	1	11
	$N_{\text{subpopulation}}$	25	0	10	35
	A allele	17	0	12	29
	a allele	33	0	8	41
	A allele frequency (p)	0.340	/	0.600	0.414
	a allele frequency (q)	0.660	/	0.400	0.586
De11	AA	4	0	0	5
	Aa	5	0	0	5
	aa	5	0	0	5
	$N_{\text{subpopulation}}$	14	0	0	15
	A allele	13	0	0	15
	a allele	15	0	0	15
	A allele frequency (p)	0.464	/	/	1.000
	a allele frequency (q)	0.536	/	/	0.000

Table 3 shows that the H_s , H_t , and F_{ST} values for five primers are calculated using allele frequencies. Sample calculations are illustrated: * $H_s = \frac{\sum_{i=1}^n 2pq}{n} = \frac{2 \times (0.507 \times 0.493 + 0.580 \times 0.420 + 0.500 \times 0.500)}{3} = 0.4957$; ** $H_t = 2\bar{p}\bar{q} = 2 \times \left(\frac{0.507 + 0.580 + 0.500}{3} \right) \times \left(\frac{0.493 + 0.420 + 0.500}{3} \right) = 0.4983$; *** $F_{ST} = \frac{H_t - H_s}{H_t} = \frac{0.4983 - 0.4957}{0.4983} = 0.0053$.

Primer	H_s	H_t	F_{ST}
Ne2	0.4957 *	0.4983 *	0.0053 ***
De7	0.3292	0.4176	0.2117
De9	0.4990	0.4995	0.0010
De10	0.4644	0.4982	0.0678
De11	0.2487	0.3922	0.3659

Table 4 HWE and Chi-square value calculated using the equation: $\chi^2 = \sum \frac{(O-E)^2}{E}$. The critical value is 3.841 ($p = 0.05$, $df=1$).

Gene	Genotype	HW ratio	Expected genotype frequency	Expected no. of genotype	Observed no. of genotype	χ^2
Ne2	AA	p^2	0.276	27	13	7.50
	Aa	$2pq$	0.499	49	78	16.60
	aa	q^2	0.225	22	8	9.18
	Total		1.000	99	99	33.28
De7	AA	p^2	0.201	17	10	3.20
	Aa	$2pq$	0.495	43	58	5.20
	aa	q^2	0.304	26	19	2.11
	Total		1.000	87	87	10.52
De9	AA	p^2	0.221	4	4	0.01
	Aa	$2pq$	0.498	8	8	0.03
	aa	q^2	0.280	5	5	0.01
	Total		1.000	17	17	0.05
De10	AA	p^2	0.172	6	5	0.17
	Aa	$2pq$	0.485	17	19	0.24
	aa	q^2	0.343	12	11	0.08
	Total		1.000	35	35	0.49
De11	AA	p^2	0.250	4	5	0.42
	Aa	$2pq$	0.500	8	5	0.83
	aa	q^2	0.250	4	5	0.42
	Total		1.000	15	15	1.67

Discussion

Gel image

The absence of solo-LTRs and preintegration sites in my gel image might be due to errors in the experiment, such as inaccurate sample preparation, inadequate protein concentration, insufficient electrophoresis conditions, or issues with the gel or buffer. Additionally, the homozygous absence of the solo-LTR in De7 could be influenced by natural genetic diversity or evolutionary processes such as genetic drift and selection.

Deviations from HWE

Deviations from HWE may not result from genotyping errors. Still, they might be led by chance events or genetic factors, encompassing a heterozygous advantage, population admixture or substructure, inbreeding, copy number variants, or small sample sizes (Cockerham, 1967, 1969; Crow, 2017; Deng, Chen & Recker, 2001; Weir et al., 2004). Higher observed heterozygotes in Ne2, De7, and De10 suggest positive selection, while lower observed heterozygotes in De11 indicate negative selection. The presence of selection challenges the assumption of HWE.

Solo-LTR and preintegration sites for Viral fingerprinting

Solo-LTRs and preintegration sites act as potential viral fingerprinting markers, enabling individuals to exhibit a unique viral combination amidst insertional polymorphisms. Statistically, identifying all individuals globally requires 21 endogenous viral elements (EVEs), as per $S^N = C$, with S (possible states) at 3 for distinct genotypes and C (theoretically identifiable individuals) at 7.888×10^9 . Additionally, F_{ST} values in Table 3 unveil varied differentiation levels among sub-populations, suggesting an uneven solo-LTR distribution for potential viral fingerprinting. However, it relies on within-population diversity, complicating interpretation and comparability, especially across species with differing effective population sizes and mutation rates (Meirmans & Hedrick, 2011).

Viral fingerprinting vs micro-satellites

Microsatellites, with repeat numbers varying in sequences of two to five bases (Dib et al., 1996), often demand polyacrylamide gel for a two-base pair resolution. However, due to polyacrylamide gel's drawbacks, agarose gel is preferred for viral fingerprinting because of its safety and resilience (Stellwagen, 2009). Despite its advantages over

microsatellites, viral fingerprinting has limitations. While a single microsatellite can distinguish numerous individuals, a solo-LTR typically differentiates only three individuals, limiting its applicability. Belshaw et al. found that, among 113 EVEs, only eight to eleven are insertionally polymorphic, impacting the utility of viral fingerprinting. Further research is needed to identify suitable EVEs, making this method advantageous when complementing inconclusive microsatellite screening or providing additional evidence.

Conclusion

This study on HERV-K family analysis explores human population genetics through ERV detection. By using PCR and gel electrophoresis, it decodes solo-LTRs, contributing to viral fingerprinting. The absence of solo-LTRs in the gel suggests potential issues in sample preparation or genetic diversity. HWE deviations in certain ERVs imply genetic factors rather than genotyping errors. Solo-LTRs and preintegration sites emerge as viral fingerprinting markers, indicating varied sub-population differentiation. However, the dependence of F_{ST} on within-population diversity challenges interpretation and comparability. While presenting advantages over microsatellites, viral fingerprinting requires further research for widespread applicability.

Word counts: 1288

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FINAL GRADE

GENERAL COMMENTS

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PAGE 1

PAGE 2



Comment 1

excellent title



Comment 2

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Comment 3

HML2



Comment 4

first identified in the genomes of the archaic humans



Comment 5

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Comment 11

V good abstract-could be a little more specific in places with some rewording



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Your experiment could not be reproduced from your methods section-more details are needed in places



Comment 27

in-silico PCR-but OK I think



Comment 28

v good

PAGE 3



Comment 29

nice table-v clear



Comment 30

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Comment 31

excellent figure showing attention to detail in the labelling



Comment 32

excellent ref! :)



Comment 33

good descriptive text here

PAGE 4



Comment 34

excellent footnotes usually go below the table though-not sure what the P-values were calculated from though



Comment 35

no comment



Comment 36

Fst doesn't have a P-value. In any case 0.0053 is v low differentiation

PAGE 5



Comment 37

v good



Comment 38

you need to state what they are again



Comment 39

excellent



Comment 40

Ok but I am still unsure what the P-values are for?-you need to be more explicit

PAGE 6



Comment 41

this is incorrect-there are more than 50 known now-you missed the papers in your lit review

PAGE 7



Comment 42

excellent lit review but you did miss some papers

YUKI WU

CRITERIA MET

How does this work meet the criteria for the grade it has been given?



Excellent write up for the most part. Excellent presentation of results as well as data handling and some excellent discussion. Neat and v good layout

AREAS TO IMPROVE

How could this work – and similar work in future – be improved, to gain a higher mark?



I have no idea what your P-values were for. I have given you the benefit of the doubt but you need to be more specific in the future. Some expected citations missing and your methods were incomplete. Other comments in the text